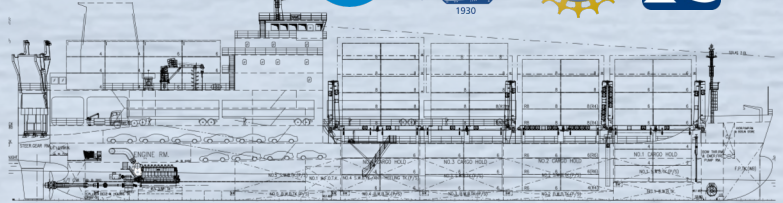


The far-reaching Scientific Impact of the NSF-OCE PO funded Oleander Project



Since the installation of an ADCP on *MV Oleander* in 1992, the **66 scientific publications** that directly use Oleander-measured velocity profiles (ADCP), temperature profiles (XBTs), and near-surface properties (TSG) have accumulated **more than 3500 citations**. These publications are authored by **159 unique researchers** (35 of whom contributed to more than one Oleander paper) and cover themes encompassing Gulf Stream transport, Atlantic Meridional Overturning Circulation, Middle Atlantic Bight shelf and slope circulation and temperature, satellite and model validation, marine heatwaves, warm and cold core rings, submesoscale flows, and cross disciplinary studies. Three of these scientific papers are accompanied by Oleander data products published on Zenodo that have been **downloaded 622** times. Oleander data underpin student research in 4 Doctoral Dissertations, 3 Master's Theses, and 3 NSF-funded Research Experiences for Undergraduates. Seminal publications include top-cited papers:

ADCP technique paper

Flagg et al. (1998). Operating an acoustic Doppler current profiler (ADCP) aboard a container vessel. Citations: **70**

Satellite velocity validation

Cornillon and Park (2001). Warm core ring velocities inferred from NSCAT. Citations: **132**

MAB Shelfbreak Jet and Slope Jet

Flagg et al. (2006). A study of the currents of the outer shelf and upper slope from a decade of shipboard ADCP observations in the MAB. Citations: **103**

Wavenumber spectra

Wang et al. (2010). Wavenumber spectrum in the Gulf Stream from shipboard ADCP observations and comparison with altimetry measurements. Citations: **96**

Gulf Stream transport

Rossby et al. (2014). On the long-term stability of the Gulf Stream transport based on 20 years of direct measurements. Citations: **127**

Validation of satellite salinity

Reul et al. (2014). Sea surface salinity structure of the meandering Gulf Stream revealed by SMOS sensor. Citations: **89**

Temperature front detection algorithms

Ullman and Cornillon (2014). Evaluation of front detection methods for satellite-derived SST data using in situ observations. Citations: **136**

Submesoscale turbulence

Callies et al. (2015). Seasonality in submesoscale turbulence. Citations: **468**

Marine heatwaves

Gawarkiewicz et al. (2019). Characteristics of an advective marine heatwave in the MAB in early 2017. Citations: **81**

1977 • NOAA/NFSC initiates XBT sampling on *MV Oleander* using ship riders.

1992 • ADCP and TSG installed on *MV Oleander*.

Timeline of Papers & Products*

1998 2
1999
2000 2
2001 3
2002
2003 1
2004 2
2005 2
2006 1
2007 1
2008 2
2009
2010 3
2011 1
2012
2013 1
2014 7
2015 3
2016 3
2017 2
2018 5
2019 4
2020 3, 2
2021 4
2022 4
2023 3
2024 2
2025 4
2026 1, 1

Gulf Stream position

Rossby and Benway (2000). Slow variations in mean path of the Gulf Stream east of Cape Hatteras. Citations: **143**

Sea Surface Salinity

Wilson et al. (2001). PALS microwave sensor for ocean salinity and soil moisture measurements. Citations: **176**

MAB primary production

Mouw and Yoder (2005). Primary production calculations in the MAB. Citations: **77**

Validation of satellite-derived velocity and EKE

Scharffenberg et al. (2010). Seasonal variations of the large-scale geostrophic flow field and eddy kinetic energy inferred from the TOPEX/Poseidon and Jason-1 tandem mission data. Citations: **105**

US Northeast shelf temperatures

Chen et al. (2014). Diagnosing the warming of the Northeastern US coastal ocean. Citations: **189**

2012 • **AXIS installed on *MV Oleander* for automated XBT sampling.**

Accelerated warming

Forsyth et al. (2015). Recent accelerated warming of the continental shelf off New Jersey: Observations from the CMV Oleander XBT line. Citations: **116**

Unbalanced submesoscale flow

Cao et al., (2023). Towards the upper-ocean unbalanced submesoscale motions in Oleander observations. Citations: **14**

*full reference list available upon request mandres@whoi.edu

Future innovations using Oleander data will include studies of unbalanced flow, comparison with SWOT satellite observations, air-sea coupling over strong fronts, data-assimilation techniques, discovering flow patterns with AI and ML, bio-physical interactions, physics and biogeochemistry of the carbon cycle, and inferring biomass from ADCP backscatter.

NSF funding for Oleander data collection is presently OCE-2241601 and has included -9819724, -0241654, -0825845, -1536851, and -1536586, with XBT probes supplied by NOAA AOML and in-kind support provided by Bermuda Container Line.

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